

ARYAN NARANG

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Backend-focused full-stack engineer building scalable systems with Node.js, TypeScript, Redis, RabbitMQ, PostgreSQL, Docker, and Next.js. Experienced in REST APIs, async job queues, real-time applications, transactional workflows, and worker-based processing with a focus on performance, reliability, and clean system design.

TECHNICAL SKILLS

- **Languages:** C++, JavaScript, TypeScript, SQL
- **Backend & Databases:** Node.js, Express.js, REST APIs, RabbitMQ, BullMQ, Redis, PostgreSQL, MongoDB
- **Systems & Auth:** Distributed Systems, Async Processing, Workers, Rate Limiting, JWT, OAuth
- **Frontend & Tools:** React.js, Next.js, Tailwind CSS, Socket.IO, WebRTC, Clerk, Docker, Git, GitHub, Vercel, AWS
- **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP

WORK EXPERIENCE

SDE Intern - ITJOBXS

May 2025 – July 2025

- Built authentication and registration workflows with secure request validation, backend verification, and database-backed access control to reduce automated registrations.
- Optimized **SQL queries** and engineered backend APIs for a **MERN-based e-commerce system**, improving request reliability and scalable authentication handling.
- Integrated **Google reCAPTCHA**, domain-based email validation, and hash-based whitelisting mechanisms to mitigate bot activity and strengthen platform security.

PROJECTS

PullVault - Transactional Marketplace & Auction Platform | Next.js | TypeScript | PostgreSQL | Redis | Socket.IO

- Built a full-stack transactional marketplace for pack purchases, peer-to-peer listings, and live auctions using **Next.js**, **PostgreSQL**, **Redis**, and **Socket.IO**, with balance holds, inventory constraints, and append-only ledger entries.
- Designed concurrency-safe purchase, listing, and bidding flows using conditional SQL updates, transaction boundaries, optimistic concurrency, and worker-based auction settlement to prevent overselling and double-spends.
- Engineered real-time auction infrastructure with **Redis pub/sub**, Socket.IO room updates, anti-snipe timer extensions, sealed bids, and automated settlement workflows.
- Added abuse-prevention and monitoring layers including Redis sliding-window rate limits, deterministic purchase jitter, wash-trade flags, and admin dashboards for marketplace health.

Distributed Job Queue System | Node.js | Express.js | RabbitMQ | Redis | PostgreSQL | Docker

- Designed an async job-processing architecture using **Node.js**, **RabbitMQ**, **Redis**, **PostgreSQL**, and **Docker** to move long-running tasks outside the API request lifecycle.
- Implemented producer-worker separation with retries, priority queues, and failure logging, enabling background jobs to run independently from user-facing API requests.
- Used **Redis-based rate limiting** to control request bursts and protect workers from overload during high-volume job submissions.
- Built **REST APIs** for job creation, status tracking, retry visibility, and worker-result retrieval, improving observability and fault diagnosis across distributed services.

YOOM - Real-Time Video Conferencing Platform | Next.js | TypeScript | Stream Video SDK | Clerk | Tailwind | Vercel

- Built a real-time video conferencing platform with meeting scheduling, instant joins, screen sharing, recordings, and dynamic meeting layouts using **Next.js** and **TypeScript**.
- Implemented **Clerk**-based authentication and role-aware meeting flows for hosts, participants, and scheduled sessions.
- Designed reusable meeting components for lobby, active calls, recordings, and scheduled meetings to improve maintainability.

EDUCATION

Vellore Institute of Technology

2022 – 2026

B.Tech in Computer Science & Engineering

ACHIEVEMENTS & CERTIFICATIONS

- Achieved **Global Rank 3,561** in a [LeetCode Weekly Contest](#), demonstrating strong algorithmic problem solving.
- Solved **300+ Data Structures & Algorithms problems** across LeetCode and other coding platforms.
- **Microsoft Certified: Azure AI Engineer Associate** - Microsoft